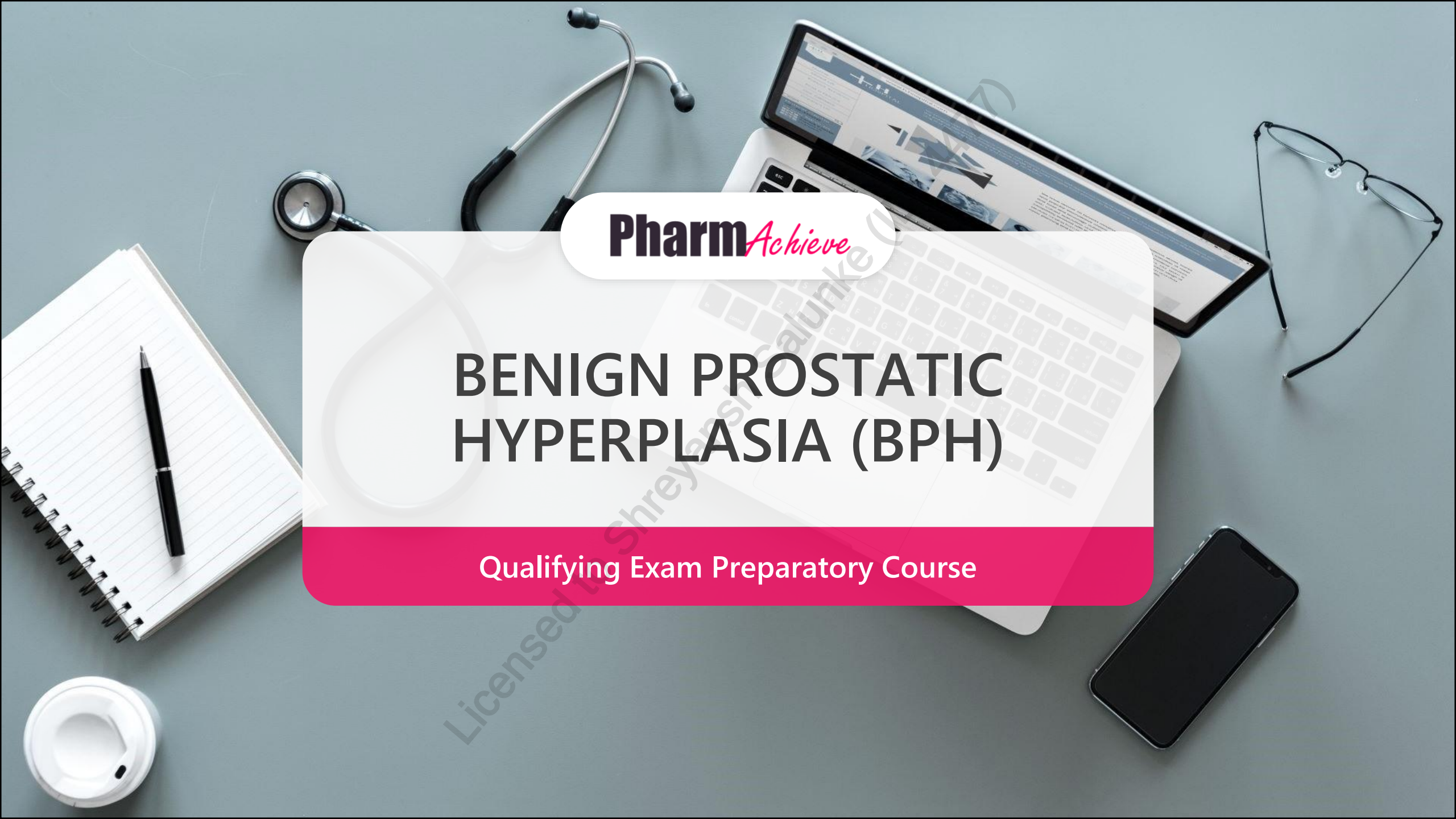




Pharm*Achieve*

BENIGN PROSTATIC HYPERPLASIA (BPH)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AE Adverse Effects

ARI Alpha Reductase Inhibitor

BPH Benign Prostatic Hyperplasia

CrCl Creatinine Clearance

DHT Dihydrotestosterone

LUTS Lower Urinary Tract Symptoms

MLUTS Male Lower Urinary Tract Symptoms

MOA Mechanism of Action

PSA Prostate Specific Antigen

PVR Post-Void Residual

R/O Rule Out

TURP Transurethral Resection of the Prostate

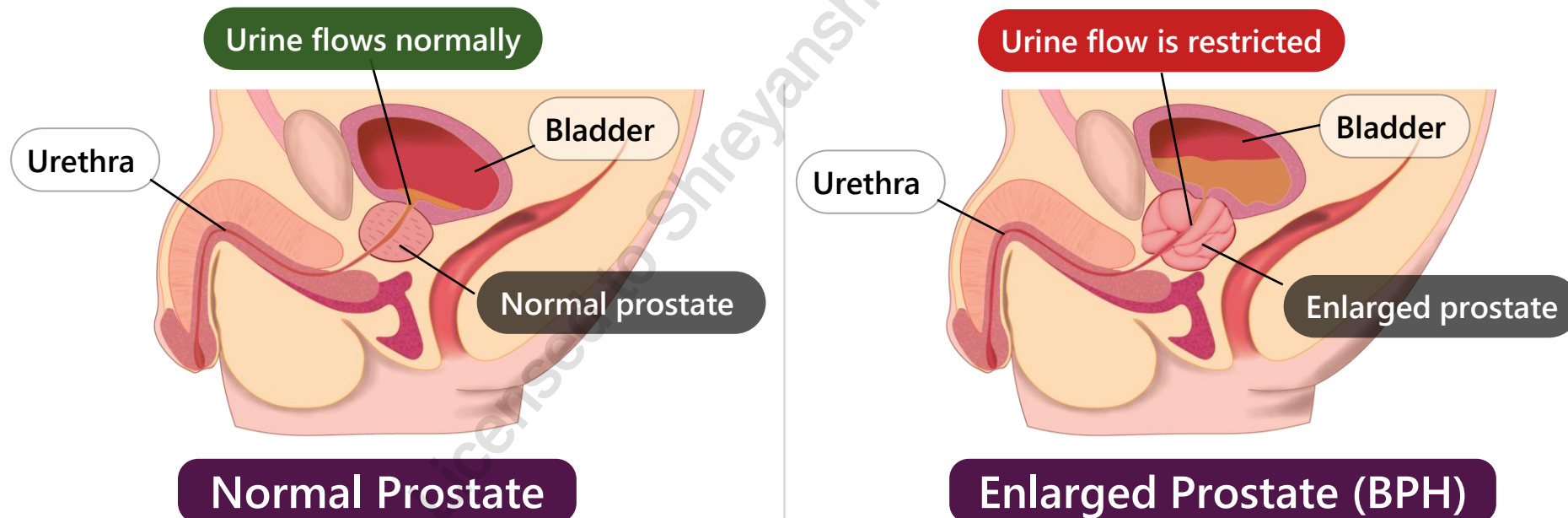
OBJECTIVES

- 1 Review the pathophysiology of BPH
- 2 Describe signs and symptoms of BPH or lower urinary tract symptoms (LUTS)
- 3 Identify requirements for diagnosis
- 4 Discuss therapies used for BPH management

DEFINITIONS

BPH is a **non-cancerous enlargement of the prostate** that can cause **urinary symptoms** identical to prostate cancer

- Common in older men
- Can cause an **elevated PSA** (prostate specific antigen) level
- BPH is **not cancer** and **cannot spread to other parts of the body**



PATHOPHYSIOLOGY

- Prostatic enlargement depends on the potent androgen **dihydrotestosterone (DHT)**
- Type II 5-alpha-reductase metabolizes circulating testosterone into **DHT**, which works locally
- **DHT binds to androgen receptors** in the cell nuclei potentially resulting in BPH
- Microscopically, BPH is characterized as a **hyperplastic process**

There are at least 3 components which contribute to the symptoms of BPH

1. **Mechanical obstruction** by the enlarged prostate
2. **Dynamic obstruction** caused by the tone of the prostatic smooth muscle
3. The **reaction of the bladder** to the obstruction

CLINICAL PRESENTATION

- Symptoms often attributed to BPH can be caused by other disease processes
 - A history and physical examination are essential in ruling out other etiologies of lower urinary tract symptoms (LUTS)
1. **Urinary frequency** - The **need to urinate frequently** during the day or night (nocturia), usually voiding only small amounts of urine with each episode
 2. **Urinary urgency** - The sudden, **urgent need to urinate**, owing to the sensation of imminent loss of urine without control
 3. **Hesitancy** - **Difficulty initiating** the urinary stream; interrupted, weak stream
 4. **Incomplete bladder emptying** - The **feeling of persistent residual urine**, regardless of the frequency of urination
 5. **Straining** - The **need to strain or push** (Valsalva maneuver) to initiate and maintain urination in order to more fully evacuate the bladder
 6. **Decreased force of stream** - The subjective **loss of force of the urinary stream** over time
 7. **Dribbling** - The **loss of small amounts of urine** due to a poor urinary stream

RISK FACTORS

1. Aging

- Not common in men < 40 years of age

2. Family history

- Having a close male relative (father or brother) with BPH

3. Medications (mimic or exacerbate symptoms)

- Antihistamines
- Diuretics

4. African descent males

5. Lifestyle

- Obesity increases risk

6. Comorbidities

- Diabetes and heart disease can increase risk

DIAGNOSTIC EVALUATION

HISTORY

- Evaluate **symptom severity and quality of life** (**Recommended** to use International Prostate Symptom Score or American Urology Association Score) with a focus on voiding & storage symptoms, onset and progression of LUTS
- **Medical history** (medications, relevant prior and current illnesses, relevant prior surgeries and trauma)

FOCUSED PHYSICAL EXAM

- **Digital rectal exam** (inaccurate for size, but can detect shape and consistency)

LABORATORY TESTS

- **Urinalysis (and culture if pyuria)** to r/o diagnoses other than BPH
- **Prostate Specific Antigen** (may be useful marker of prostate size and may predict risk of BPH progression)

OPTIONAL

(reasonable if diagnostic uncertainty or physician feels it is indicated)

- Serum creatinine
- Uroflowmetry
- Post-void residual
- Voiding diary

NOT RECOMMENDED

(may be required in patients with definite indication)

- Urine cytology
- Cystoscopy
- Urodynamics
- Radiological evaluation of upper urinary tract
- Biopsy
- Prostate ultrasound

DIFFERENTIAL DIAGNOSIS AND DRUG RELATED CAUSES

DIFFERENTIAL DIAGNOSIS

- Bladder cancer*
- Prostate cancer*
- Prostatitis
- Bladder stones
- Interstitial cystitis
- Radiation cystitis
- Urinary tract infection
- Diabetes mellitus
- Parkinson's disease
- Primary bladder neck hypertrophy
- Congestive heart failure
- Lumbosacral disc disease
- Multiple sclerosis
- Nocturnal polyuria

DRUG RELATED CAUSES

- **Antihistamines**
 - Prevent bladder muscle from contracting (decrease parasympathetic tone)
- **Decongestants**
 - Tighten muscles in prostate and bladder neck making it more difficult for urine to leave bladder
- **Diuretics**
 - Increased urine production
- **Opiates**
 - Impaired bladder contractility
- **Tricyclic antidepressants**
 - Anticholinergic effects

* A urological consultation is warranted if cancer is suspected, medical therapy is not effective, and/or complications are present

GOALS OF THERAPY

- ✓ Provide relief or resolution of lower urinary tract symptoms (LUTS)
- ✓ Decrease risk of progression of BPH
- ✓ Prevent complications from bladder outlet obstruction
- ✓ Early diagnosis of prostate cancer
- ✓ Reduce risk of surgical intervention

NONPHARMACOLOGIC THERAPY

- **Watchful waiting**
 - Periodic visits to physician – no pharmacologic treatment
- **Lifestyle changes**
 - Fluid restriction
 - Avoidance of irritative beverages (e.g. caffeine or alcohol)
 - Avoidance and/or monitoring of some medications (e.g. diuretics, decongestants, antihistamines, antidepressants)
 - Elevate legs prior to resting if existing pedal edema
 - Timed voiding (bladder retraining)
 - Pelvic floor exercises (in some cases)
 - Treat constipation
- **Surgery**
 - TURP, open simple prostatectomy, laser prostatectomy, transurethral incision of prostate, minimally invasive techniques
 - Follow-up 4-6 weeks post-op for catheter removal and to evaluate treatment response

PHARMACOLOGICAL THERAPY

Selective Alpha-Blockers

Drugs:	Silodosin, Alfuzosin, Tamsulosin
MOA:	Block alpha-adrenergic receptors that mediate smooth muscle activity in the bladder neck, prostate and prostatic capsule. Relaxes the smooth muscle to allow for improved urinary flow rates. Provides symptomatic relief
Efficacy:	Exerts effect quickly, does not decrease risk of progression (little impact on prostate growth, risk of urinary retention or need for BPH surgery) All 3 agents appear to have equal clinical effectiveness
Duration:	Effects can be evaluated as early as 1 week, may take up to 6 weeks for full effect
Interactions:	CYP 3A4 Inhibitors
AEs:	Tamsulosin: intraoperative floppy iris syndrome (does not appear to be an issue in men with no planned cataract surgery), cataract formation, retrograde ejaculation Silodosin: retrograde ejaculation (quite common) Dizziness, headaches, asthenia, nasal congestion, hypotension (less common than non-selective)
Tips:	Avoid silodosin: CrCl <30 mL/min (reduced dose if 30-50 mL/min) Administer silodosin & alfuzosin with food, tamsulosin with or without food Rare cases of allergic reaction to tamsulosin in patients with sulfa allergies

PHARMACOLOGICAL THERAPY

Non-Selective Alpha-Blockers

Drugs:	Doxazosin, Terazosin → Nonselective (require dose titration and blood pressure monitoring)
MOA:	Block alpha-adrenergic receptors that mediate smooth muscle activity in the bladder neck, prostate and prostatic capsule. Relaxes the smooth muscle to allow for improved urinary flow rates. Provides symptomatic relief
Efficacy:	Exerts effect quickly, does not decrease risk of progression (little impact on prostate growth, risk of urinary retention or need for BPH surgery) Both agents appear to have equal clinical effectiveness
Duration:	Effects can be evaluated as early as 1 week, may take up to 6 weeks for full effect
Interactions:	Antihypertensives (increased hypotension), CYP 3A4 Inhibitors
AEs:	Dizziness (quite common), hypotension (quite common), headaches, asthenia, nasal congestion
Tips:	Minimize side effects by taking at bedtime

PHARMACOLOGICAL THERAPY

5-Alpha Reductase Inhibitors

Drugs:	Finasteride, Dutasteride
MOA:	Blocks the conversion of testosterone to dihydrotestosterone (DHT), androgen believed to be responsible for prostate enlargement. <i>Modest 20-30% shrinkage of prostate size.</i>
Efficacy:	6-12 months for symptomatic relief, ↓ PSA by 50%; ↓ risk of progression by reducing prostate size and therefore decrease the risk of surgical intervention, efficacy noted in patients with prostate volume >30 mL
Duration:	If effective, lifelong therapy otherwise prostate will re-grow to original size
Interactions:	CYP3A4 Inhibitors (dutasteride); No clinically significant interactions (finasteride)
AEs:	Impotence, sexual dysfunction, ↓ libido, ↓ semen quantity at ejaculation, gynecomastia (rare); risk of suicidal ideation (rare; finasteride)
Tips:	• Drug compliance is important for effective decrease in PSA levels • Rise in PSA after 6 months should question compliance or send for urology referral • Require hazardous drug handling considerations • Reduce risk of prostate cancer (not approved for prevention in Canada)

PHARMACOLOGICAL THERAPY

Phosphodiesterase Type 5 Inhibitors

Drugs:	Tadalafil
MOA:	Relaxes prostatic smooth muscle
Efficacy:	Quick onset
Indication:	Daily management of erectile dysfunction and treatment of the signs and symptoms of BPH
Interactions:	Alpha blockers, nitrate-based medication (life-threatening hypotension)
AEs:	Dyspepsia, headache, nasal congestion, back pain, flushing, visual disturbances, permanent vision or hearing loss (rare)
Tips:	• CrCl <30 mL/min: Avoid once-daily dosing • Avoid use with nitrates → Decrease in blood pressure • May be administered with or without food

PHARMACOLOGICAL THERAPY

Antimuscarinics

Drugs:	Fesoterodine, Oxybutynin, Solifenacin, Tolterodine
MOA:	Binds to muscarinic cholinergic receptors (acts as a competitive antagonist at these receptors)
Efficacy:	Quick onset
Indication:	Symptomatic relief of overactive bladder – helpful for storage symptoms (e.g. urgency, frequency, and nocturia)
Interactions	Tricyclic antidepressants, anticholinergics
AEs:	Dry mouth, drowsiness, constipation Small risk of urinary retention – tends to occur during first month (tolterodine, fesoterodine and mirabegron shown to have low rates of urinary retention)
Tips:	- Used in combination with alpha-blockers (may be useful in men experiencing both voiding and storage symptoms who have failed alpha blocker monotherapy) - May need to try multiple agents before finding “the right one” - Do not use in patients with uncontrolled narrow-angle glaucoma - Risk of urinary retention – use with caution if residual urine volume >250 cc (lack of evidence for safety in men with high post-void residual (PVR) volume)

PHARMACOLOGICAL THERAPY

Beta-3 Adrenergic Agonists

Drugs:	Mirabegron ER
MOA:	Potent, selective beta-3 adrenergic agonist that relaxes bladder smooth muscle and reduces voiding frequency
Indication:	Symptomatic relief of overactive bladder
Interactions:	Mirabegron is a moderate inhibitor of CYP2D6 and weak inhibitor of P-gp → monitor interactions
AEs:	Hypertension, nasopharyngitis, urinary tract infection, tachycardia
Tips:	- Fairly well tolerated compared to the antimuscarinics - Has been studied in combination with antimuscarinic therapy for incontinence - Avoid in severe, uncontrolled hypertension - Should not be chewed or crushed

PHARMACOLOGICAL THERAPY

Desmopressin

Desmopressin

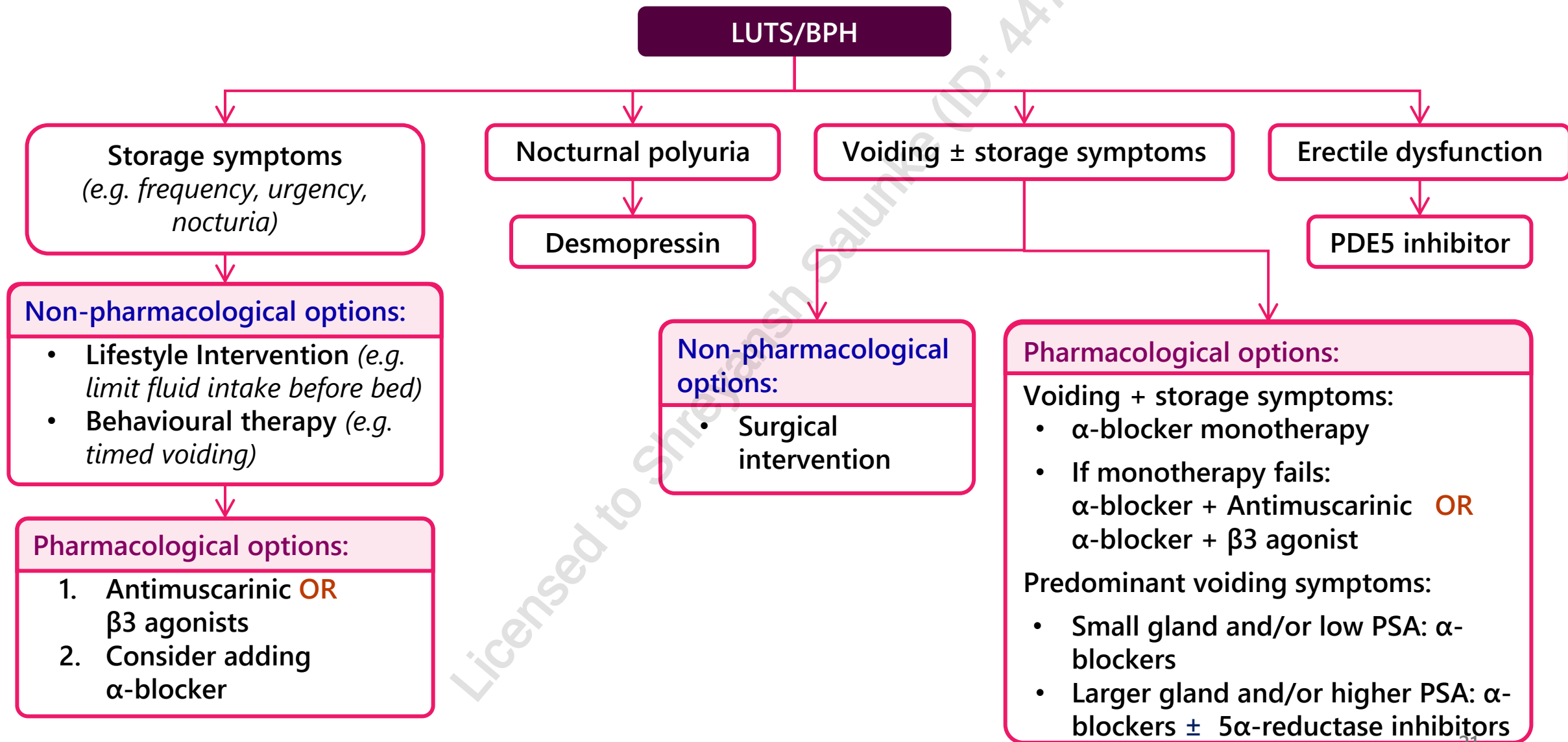
Drugs:	Desmopressin
MOA:	Synthetic analogue of the antidiuretic hormone, vasopressin. Reduces urine production by increasing water reabsorption in the kidney. Useful in men with <i>nocturnal polyuria</i>
Efficacy:	Quick onset (onset: 1 hour), reduces number of nocturnal voids and increases hours of undisturbed sleep
Duration:	Oral, sublingual tablet: 6-14 hours
Interactions:	Loop diuretics, corticosteroids, any hyponatremia-associated agents
AEs:	Hyponatremia, xerostomia (dry mouth), headache, dizziness, abdominal pain
Tips:	- Sodium levels must be taken at baseline in all men, followed by 7 days and 30 days after initiation of therapy - Sodium levels should be taken with any dose increase, as well as periodically during treatment

PHARMACOLOGICAL THERAPY

Combination Therapy: Alpha-Blockers + 5 Alpha-Reductase Inhibitors

- Appropriate and effective treatment strategy for patients **with LUTS associated with prostate enlargement (>30 mL)**
- Rapid relief of symptoms by alpha-blocker + more sustained relief of symptoms by 5 alpha-reductase inhibitor by reducing PSA, prostate volume, altering disease progression & possibly prevent need for surgery
- Downside is **additive side effects** of dual therapy
 - Increased association with new cardiac failure (combination therapy or alone-highest risk for nonselective alpha-blockers)
- Provides the **most improvement in LUTS** (improve symptoms & flow rate) compared to either agents alone
- Patients successfully treated with combination therapy may be given the option of **discontinuing the alpha-blocker after 6 to 12 months of therapy**
- Advantage
 - Lower cost & fewer side effects compared to surgical interventions and better compliance
- If **symptoms re-occur, the alpha-blocker should be restarted**

CUA 2022 GUIDELINE UPDATE



MONITORING PARAMETERS

Efficacy & Safety Endpoints		
Parameter	Degree of Change	Timeframe
LUTS/ AUA score	Improved	1 week (Alpha blocker) 6 months (5ARIs)
ADRs (headache, alopecia, impotence, premature ejaculation, dizziness, etc.)	Prevent	During course of therapy
PSA	Improvement	Controversial

TAKEAWAY

- **Alpha-blockers** work quickly (4 weeks) but **do not decrease the risk of progression**
 - Silodosin, Alfuzosin, Tamsulosin → **Selective**
 - Doxazosin, Terazosin → **Nonselective** (dose titration & blood pressure monitoring required)
- **5-ARIs** work *slowly* (6-12 months); **decrease the risk of progression** by ↓ PSA & prostate size
- **5-ARIs** work better in men with large prostates (>30 mL) and higher PSA levels (>1.4 ng/mL)
- Combination therapy of alpha-blocker and 5-ARI works better than monotherapy → may be used for patients with large prostates or high PSA level with moderate-severe symptoms
 - If successful, can stop the alpha-blocker after 6 to 12 months of therapy
 - If symptoms re-occur, the alpha-blocker should be restarted
- **Combination therapies:**
 1. **Alpha-blocker + anti-muscarinic** OR **alpha-blocker + β 3 agonist** → for mixed LUTS/BPH in men after failure of alpha-blocker monotherapy
 2. **Alpha-blocker + 5-ARI** → symptomatic LUTS and prostate size > 30 mL
- Tadalafil: erectile dysfunction & BPH; can't give with alpha-blocker
- Current guidelines recommend **against** the use of phytotherapies (e.g. saw palmetto) since no significant difference versus placebo in numerous studies

CASE 1

CD is a 62-year-old male who presents to your clinic with complaints of urinary frequency and incomplete voiding. He states these symptoms are beginning to affect his performance at work, as he feels the need to urinate at least 8 times during his workday. His medical history is significant for dyslipidemia, hypertension and restless legs syndrome. He takes rosuvastatin 10 mg po daily, candesartan 4 mg po daily, vitamin D 1000 units po daily, magnesium oxide 420 mg po qhs and saw palmetto 450 mg po daily. He has no known drug allergies. He drinks 3-5 alcoholic drinks per week. His blood pressure in the office today was 122/85 mmHg.

1. Which of the following represents the most appropriate next step?
 - a) Recommend dual therapy with tamsulosin + finasteride. Schedule a follow-up appointment with CD in 6 months. If symptoms have improved, tamsulosin can be stopped
 - b) Advise CD to stop drinking alcohol. Follow up in 3 months to see if symptoms have improved.
 - c) Recommend CD be seen for a digital rectal exam, urinalysis and PSA level
 - d) Recommend tamsulosin only, as there is no evidence that CD's prostate is significantly enlarged



CASE 1

CD is a 62-year-old male who presents to your clinic with complaints of urinary frequency and incomplete voiding. He states these symptoms are beginning to affect his performance at work, as he feels the need to urinate at least 8 times during his workday. His medical history is significant for dyslipidemia, hypertension and restless legs syndrome. He takes rosuvastatin 10 mg po daily, candesartan 4 mg po daily, vitamin D 1000 units po daily, magnesium oxide 420 mg po qhs and saw palmetto 450 mg po daily. He has no known drug allergies. He drinks 3-5 alcoholic drinks per week. His blood pressure in the office today was 122/85 mmHg.

Test results came back, and CD's PSA is 1.2 ng/mL. His prostate size is 22 mL.

2. Which of the following now represents the most appropriate next step?
- a) Recommend dual therapy with tamsulosin + finasteride. Schedule a f/u appointment with CD in 6 months. If symptoms have improved, tamsulosin can be stopped
 - b) Recommend tamsulosin only, as CD's prostate volume is < 30 mL so a 5A reductase inhibitor is not indicated at this time
 - c) Recommend tadalafil only, as CD's prostate volume is < 30 mL so a 5A reductase inhibitor is not indicated at this time
 - d) Recommend lifestyle modifications and watchful waiting, as CD's symptoms are mild



CASE 1

CD is a 62-year-old male who presents to your clinic with complaints of urinary frequency and incomplete voiding. He states these symptoms are beginning to affect his performance at work, as he feels the need to urinate at least 8 times during his workday. His medical history is significant for dyslipidemia, hypertension and restless legs syndrome. He takes rosuvastatin 10 mg po daily, candesartan 4 mg po daily, vitamin D 1000 units po daily, magnesium oxide 420 mg po qhs and saw palmetto 450 mg po daily. He has no known drug allergies. He drinks 3-5 alcoholic drinks per week. His blood pressure in the office today was 122/85 mmHg.

Test results came back, and CD's PSA is 1.2 ng/mL. His prostate size is 22 mL.

3. Which of the following statements is NOT a correct counselling point for alpha blockers?
- a) Non-selective alpha blockers can be taken at bedtime to minimize adverse effects
 - b) Non-selective alpha beta require dose titration to minimize adverse effects
 - c) Tamsulosin and silodosin can be taken with or without food
 - d) It can take up to 4 weeks to see an effect



CASE 1

CD is a 62-year-old male who presents to your clinic with complaints of urinary frequency and incomplete voiding. He states these symptoms are beginning to affect his performance at work, as he feels the need to urinate at least 8 times during his workday. His medical history is significant for dyslipidemia, hypertension and restless legs syndrome. He takes rosuvastatin 10 mg po daily, candesartan 4 mg po daily, vitamin D 1000 units po daily, magnesium oxide 420 mg po qhs and saw palmetto 450 mg po daily. He has no known drug allergies. He drinks 3-5 alcoholic drinks per week. His blood pressure in the office today was 122/85 mmHg.

Test results came back, and CD's PSA is 1.2 ng/mL. His prostate size is 22 mL.

CD returns to his MD after 6 months to report that although he has seen some improvement, he is still experiencing significant symptoms. Upon further investigation, the MD notes that the PSA has gone up to 1.5 ng/mL and his prostate is enlarged (>40 mL).

4. What is the most appropriate course of action?

- a) Switch to another drug within the same class of medication
- b) Discontinue the current medication, start dutasteride
- c) Add dutasteride to his regimen
- d) Add tadalafil to the current regimen



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CHANGE LOG

August 2025

- Slide 9: Removed FYI
- Slide 13 & 14: Split up into 2 slides to make it more readable
- Slides 13-19: Removed incidences
- Slide 24, 25, 26: Updated case and questions

November 2025

- Slide 10: Added note on urological consultation
- Slide 21: Updated algorithm on management of voiding + storage symptoms as per the guidelines to indicate that alpha-blocker monotherapy should be trialled first before combination therapy
- Slide 23: Consolidated takeaway and added summary of combination therapies

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 15: Simplified mechanism of action
- Content reviewed- no changes made